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The geopolymerisation of alumino-silicate minerals

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Abstract

Geopolymers are similar to zeolites in chemical composition, but they reveal an amorphous microstructure. They form by the co-polymerisation of individual alumino and silicate species, which originate from the dissolution of silicon and aluminium containing source materials at a high pH in the presence of soluble alkali metal silicates. It has been shown before that geopolymerisation can transform a wide range of waste alumino-silicate materials into building and mining materials with excellent chemical and physical properties, such as fire and acid resistance. The geopolymerisation of 15 natural Al–Si minerals has been investigated in this paper with the aim to determine the effect of mineral properties on the compressive strength of the synthesised geopolymer. All these Al–Si minerals are to some degree soluble in concentrated alkaline solution, with in general a higher extent of dissolution in NaOH than in KOH medium. Statistical analysis revealed that framework silicates show a higher extent of dissolution in alkaline solution than the chain, sheet and ring structures. In general, minerals with a higher extent of dissolution demonstrate better compressive strength after geopolymerisation. The use of KOH instead of NaOH favours the geopolymerisation in the case of all 15 minerals. Stilbite, when conditioned in KOH solution, gives the geopolymer with the highest compressive strength (i.e., 18 MPa). It is proposed that the mechanism of mineral dissolution as well as the mechanism of geopolymerisation can be explained by ion-pair theory. This study shows that a wide range of natural Al–Si minerals could serve as potential source materials for the synthesis of geopolymers.

Keywords: Al–Si minerals; Geopolymers; Silicates

1. Introduction

Since 1978, Joseph Davidovits has developed amorphous to semi-crystalline three-dimensional alumino-silicate materials, which he called ‘‘geopolymers’’ (mineral poly-

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mers resulting from geochemistry) (Comrie and Davidovits, 1988; Davidovits, 1988a,b, 1991, 1994; Davidovits and Davidovics, 1988; Davidovits et al., 1990, 1994). Geopolymerisation involves a chemical reaction between various alumino-silicate oxides (Al^{3+} in IV–V fold coordination) with silicates under highly alkaline conditions, yielding polymeric Si–O–Al–O bonds, which can be presented schematically as follows:

$$\underset{\text{(Si-Al materials)}}{n(Si_2O_5,Al_2O_2)} + 2nSiO_2 + 4nH_2O + NaOH \text{ or } (KOH) \rightarrow Na^+,K^+ + \underset{\text{(Geopolymer precursor)}}{n(OH)_3\text{-}Si\text{-}O\text{-}\underset{\underset{(OH)_2}{|}}{Al^-}\text{-}O\text{-}Si\text{-}(OH)_3} \tag{1}$$

$$n(OH)_3\text{-}Si\text{-}O\text{-}\underset{\underset{(OH)_2}{|}}{Al^-}\text{-}O\text{-}Si\text{-}(OH)_3 + NaOH \text{ or } (KOH) \rightarrow \underset{\text{(Geopolymer backbone)}}{(Na^+,K^+)\text{-}(\text{-}\overset{|}{\underset{\underset{|}{\underset{O}{|}}}{Si}}\text{-}O\text{-}\overset{|}{\underset{\underset{|}{\underset{O}{|}}}{Al^-}}\text{-}O\text{-}\overset{|}{\underset{\underset{|}{\underset{O}{|}}}{Si}}\text{-}O\text{-})} + 4nH_2O \tag{2}$$

The above two reaction paths indicate that any Si–Al materials might become sources of geopolymerisation (Van Jaarsveld et al., 1997). According to Davidovits (1994), geopolymeric binders are the amorphous analogues of zeolites and require similar hydrothermal synthesis conditions. Reaction times, however, are substantially faster, which results in amorphous to semi-crystalline matrices compared with the highly crystalline and regular zeolitic structures. The electron diffraction analysis conducted by Van Jaarsveld et al. (1999) showed that the structure of geopolymers is amorphous to semi-amorphous. The exact mechanism by which geopolymer setting and hardening occur is not fully understood. Most proposed mechanisms consist of a dissolution, transportation or orientation, as well as a reprecipitation (polycondensation) step (Davidovits, 1988a; Van Jaarsveld et al., 1998). It appears that an alkali metal salt and/or hydroxide is required for dissolution of silica and alumina to proceed, as well as for the catalysis of the condensation reaction. In alumino-silicate structures silicon is always 4 co-ordinated, while aluminium ions can be 4 or 6 co-ordinated. It is possible that the coordination number of aluminium in the starting materials will have an effect on its eventual bonding in the matrix. A highly reactive intermediate gel phase is believed to form by co-polymerisation of individual alumino and silicate species. Little is known about the behaviour of this gel phase and the extent to which the nature of the starting materials and the actual concentrations in solution are affecting the formation and setting of this gel phase. A major experimental problem is that the gel phase cannot be ''frozen'' and then analysed to observe the evolution of its composition and texture.

It is well known that geopolymers possess excellent mechanical properties, fire resistance, and acid resistance (Davidovits and Davidovics, 1988; Palomo et al., 1992). These properties make geopolymers a potential construction material, which has attracted a great deal of attention internationally in the past 20 years (Malone et al., 1986; Laney, 1993; Davidovits et al., 1994; Van Jaarsveld et al., 1997). Although the commercial applications of geopolymers are limited at present, a recent increase in research and development activity could facilitate the wider acceptance of these materials. In previous papers, many Al–Si containing source materials such as building residues, flyash, furnace slag, pozzolan, and some pure Al–Si minerals and clays (kaolinite and metakaolinite) have been studied (Van Jaarsveld et al., 1997, 1998, 1999). In fact, some research results have already been applied successfully in industry to substitute traditional cement. Nevertheless, most of these studies have used the source materials on an arbitrary basis without consideration of the mineralogy or paragenesis of the individual minerals. This means that no generic knowledge is available on the propensity of Al–Si minerals to geopolymerise, despite the availability of some data on the solubility of selected minerals in alkaline medium. Usually, the interrelationship between mineralogy and reactivity of individual minerals is extremely complex, and this is the reason why previous studies have focused on the geopolymerisation of selected materials that are widely available. More than 65% of the crust of the earth consists of Al–Si materials, so that it is most useful to understand how individual Al–Si minerals will geopolymerise. Such information will enhance the commercialisation of this promising new technology.

The primary aim of this paper is to demonstrate that a wide range of Al–Si minerals could form geopolymers. Secondly, an attempt is made to relate the composition, physical properties, mineralogy and paragenesis of these minerals to the compressive strengths of the final synthesised matrices. A mechanism of geopolymerisation will also be proposed. Sixteen natural Al–Si minerals — which cover the ring, chain, sheet, and framework crystal structure groups, as well as the garnet, mica, clay, feldspar, sodalite, and zeolite mineral groups — were investigated. It will be shown that all these minerals, except hydroxyapophyllite, produced acceptable matrices.

2. Experimental methods

Sixteen natural Al–Si minerals were bought from ''Geological Specimen Supplies'', Turramurra, NSW, Australia and were reduced in size and sieved to -212 μm. The approximate formula for each mineral is given in Table 1, which also gives the hardness and density values. Table 1 shows the elemental composition of each of these minerals, which was obtained by X-ray Fluorescence (XRF) analysis, using a Siemens SRS 3000 instrument. X-ray Diffractograms (XRD) were recorded on a Philips PW 1800 machine to give structural information on each mineral sample and the formed geopolymer, using Cu K_α and a scanning rate of 2°/min from 6 to 65° (2θ). Hydroxyapophyllite did not give an acceptable geopolymer, so that the matrix could not be analysed.

The extent of dissolution of the 16 minerals in alkaline medium was determined by mixing 0.50 ± 0.002 g of each mineral with 20 ml of alkaline solution (2, 5, and 10 N of NaOH or KOH) at room temperature for 5 h using a magnetic stirrer. After filtration the solution part was diluted to 0.2 N alkaline concentration and neutralised by 36% HCl. A Perkin Elmer 3000 Inductively Coupled Plasma was used to analyse the filtered solutions, with scandium being used as an internal standard.

In real geopolymeric reactions, the mass ratio of alumino-silicate powder to alkaline solution is ≈ 3.0, which causes the alkaline solution to form a thick gel instantaneously

Table 1
Elemental composition and physical properties of selected alumino-silicate minerals

Mineral	Ideal stoichiometry[a]	Composition, wt.% (XRF)[b]					Hardness	Density	Contaminant	Molar Si/Al ratio
		SiO_2	Al_2O_3	M1[e]	M2[e]	M3[e]	Mohs[a,c]	$[g/cm^3]$[a,c]	(XRD)[d]	based on XRF
Ortho-, di-, and ring silicate										
Almandine	$Fe_3Al_2(SiO_4)_3$	38.57	20.09	Fe_2O_3 36.71	MnO 4.06	MgO 2.25	6.5–7.5	4.3		1.601
Grossular	$Ca_3Al_2(SiO_4)_3$	48.53	1.59	Fe_2O_3 9.68	CaO 25.41	MgO 1.26	6.5–7.5	3.55	Quartz	
Garnet group										
Sillimanite	Al_2SiO_5	40.8	57.78				6.5–7.5	3.24		0.599
Andalusite	Al_2SiO_5	39.87	43.63	K_2O 4.32	CaO 4.05		6.5–7.5	3.14	Margarite Mon. Muscovite-3T Rhom.	0.775
Kyanite	Al_2SiO_5	38.97	44.68	Fe_2O_3 4.76	K_2O 3.90	MgO 4.9	5.5–7	3.6	Zinnwaldite Mon.	0.738
Pumpellyite (Fe^{3+})	$Ca_2Fe^{3+}Al_2(SiO_4)(Si_2O_7)$-$(OH, O)_2 \cdot H_2O$	46.83	15.28	Fe_2O_3 10.95	CaO 12.59	MgO 6.31	5.0–6.0	3.35	Sepiolite Ortho.	2.598
Chain silicate										
Spodumene	$LiAlSi_2O_6$	62.84	26.58	$Fe_2O_3$1.85	K_2O 1.51	MgO 0.58	6.5–7.0	3.1		2.006
Augite	$(Ca, Mg, Fe)_2(Si, Al)_2O_6$	44.47	14.92	Fe_2O_3 12.4	CaO 6.68	MgO 10.23	5.5–6.0	3.3	Ephesite Tric.	2.526
Sheet silicate										
Lepidolite	$K(Li, Al)_3(Si, Al)_4$-$O_{10}(F, OH)_2$	49.55	28.58	K_2O 9.99			2.5–4.0	2.84		1.47
Mica group										
Illite	$(K, H_3O)Al_2(Si_3Al)$-$O_{10}(H_2O, OH)_2$	58.01	20.14	Fe_2O_3 4.93	K_2O 6.04	MgO 2.54	1.0–2.0	2.7	Quartz	2.444

Framework silicate										
Clay group										
Celsian	$BaAl_2Si_2O_8$	46.29	25.94		CaO 9.96		6.0–6.5	3.38		1.513
Feldspar group										
Sodalite	$Na_4(Si_3Al_3)O_{12}Cl$	27.57	21.51	CaO 10.76	Cl 46741 ppm	Na_2O 11.53	5.5–6.0	2.25		1.087
Sodalite group										
Hydroxyapophyllite	$KCa_4Si_8O_{20}(OH, F)\cdot 8H_2O$	51.6	0.21	K_2O 5.01	CaO 22.71		4.5–5.0	2.36		
Stilbite	$NaCa_4(Si_{27}Al_9)O_{72}\cdot 30H_2O$	58.47	15.04		CaO 7.61		3.5–4.0	2.2		3.298
Heulandite	$(Na, K, Ca, Sr, Ba)_5(Al_9Si_{27})O_{72}\cdot 26H_2O$	64.38	12.6	Fe_2O_3 6.93	CaO 2.25	Na_2O 3.63	3.5–4.0	2.2		4.338
Zeolite group										
Anorthite	$CaAl_2Si_2O_8$	46.38	14.87	Fe_2O_3 11.81	CaO 6.58	MgO 9.88	6.0–6.5	2.76	Augite	2.643

[a] Nickel and Nichols (1991).
[b] Experimental XRF results.
[c] Deer et al. (1992).
[d] Experimental XRD results.
[e] Main metal oxides contained in minerals.

upon mixing with the minerals. At that stage the dissolution reaction proceeds simultaneously with the gel formation and polycondensation (setting) reactions, so that the dissolution reaction cannot be isolated. Since the gel phase cannot be separated or analysed in situ, a dissolution procedure with lower solid/solution ratio has been chosen to investigate the dissolution behaviour of minerals. At 10 N NaOH, it becomes impractical to use filtration as a means of separating the dissolving solids from the alkaline solution at solid/solution ratios higher than 0.25. Moreover, in both NaOH and KOH solutions, it was found that the concentration of Al or Si after a certain time was linearly dependent on the solid/solution ratio, provided that sufficient excess alkali was present. Consequently, the extent of dissolution of the minerals at low solid/solution ratios could be used to predict their performance at high solid/solution ratios.

In order to achieve homogeneously mixed geopolymers and in view of the restricted availability of some mineral samples, very small samples were prepared. In all tests, 10.0 g of mineral and 5.0 g of kaolinite were dry mixed for 10 min, followed by the addition of 0.9 g of sodium silicate solution (with [Si] = 0.74 M) and 5.0 ml of 10 N KOH or NaOH solution, followed by a further 2 min of mixing by hand. The resulting slurry was then transferred to steel moulds measuring $20 \times 20 \times 20$ mm, which was followed by a gel setting and hardening stage at 35°C for 72 h. After being analysed by XRD to ensure that all samples were dried, the resulting compressive strength of each geopolymer was tested on a Tinius Tolsen compressive testing machine. It should be noted that such small samples are well below the minimum required in standard testing specifications, so that the obtained MPa values should not be interpreted in absolute but rather in relative terms. It should also be realised that such compressive strengths could be substantially higher when the reacting minerals occur in combination with filling or aggregate material of a suitable particle size distribution, similar to what happens in concrete.

The concentration of the silicate solution used in this research was [Si] = 0.74 M. The aim of adding sodium silicate solution was to enhance the formation of geopolymer precursors upon contact between a mineral and the solution. In view of the different extents of dissolution displayed by the various minerals, it is necessary to optimise the concentration of the sodium silicate solution in each case, as this concentration affects the properties of the ultimate geopolymer. Owing to the limited supply of mineral samples, such an optimisation was conducted only for stilbite by keeping all other conditions constant and using sodium silicate concentrations ranging from [Si] = 0.72 to 3.7 M. It was found that [Si] = 0.74 M, yielded optimal compressive strengths for both NaOH and KOH conditions in the case of stilbite. This concentration was then applied in the case of all other minerals without further optimisation. The concentration of sodium silicate used by other researchers ranges from [Si] = 0.72 to 3.96 M (Palomo et al., 1992; Van Jaarsveld et al., 1997, 1998, 1999).

Kaolinite and metakaolinite are relatively inexpensive alumino-silicates which have been used in most previous studies on geopolymerisation (Comrie and Davidovits, 1988; Palomo et al., 1992; Rahier et al., 1996; Van Jaarsveld et al., 1997, 1998, 1999). Many of these studies have utilised kaolinite or metakaolinite as a secondary source of soluble Si and Al in addition to waste or natural aluminosilicate materials to synthesise geopolymers. Often the rate of dissolution of Al from the waste or natural alumino-silicates is insufficient to produce a gel of the desired composition. In such cases the addition of kaolinite is necessary. However, if only kaolinite is used without the presence of other alumino-silicates a weak structure is formed, so that the synergy between different alumino-silicates seems to be important. This is an aspect that requires considerable further research. In the present study, it has been found that some of the natural alumino-silicate minerals such as stilbite and sodalite could form geopolymers on their own accord, while other weakly reactive minerals could not form acceptable bonds without the presence of kaolinite. Consequently, it has been decided to add the same amount of kaolinite to each of the minerals in order to allow a more reasonable comparison between minerals and also to allow comparison with previously published results.

3. Characterisation of minerals

The XRF analyses of 16 natural Al–Si minerals are listed in Table 1. Four crystal structure groups (ortho-, di- and ring silicates, chain silicates, sheet silicates, and framework silicates) and six mineral groups (garnet, mica, feldspars, clay, sodalite, and zeolite) are given. All 16 minerals contain SiO_2 and Al_2O_3, with the SiO_2 content varying from 27.57 wt.% in sodalite to 64.38 wt.% in heulandite. The Al_2O_3 content varies from 0.21 wt.% in hydroxyapophyllite to 57.78 wt.% in sillimanite.

The main metallic elements contained in the 16 natural minerals are Fe, Ca, Mg, K, and Na. There are nine minerals — almandine, grossular, kyanite, pumpellyite, spodumene, augite, illite, heulandite, and anorthite — that contain some iron. Among them, almandine pumpellyite, and augite contain iron in their chemical formula, while the others contained iron by paragenesis. It is known in the cement industry that Fe_2O_3, as one of five main components (Al_2O_3, SiO_2, SiO_3, CaO, Fe_2O_3), contributes to the strength development of portland cement at later ages (Popovics, 1992). In geopolymerisation, it is still an open question whether iron has any effect on strength development.

Ten minerals — grossular, andalusite, pumpellyite, augite, celsian, sodalite, hydroxyapophyllite, stilbite, heulandite, and anorthite — contain calcium with the CaO content varying from 2.25 wt.% in heulandite to 25.41 wt.% in grossular. The calcium content is an important factor affecting the quick setting and final strength in concrete (Popovics, 1992), and there are indications that it may also affect the properties of geopolymers (Davidovits, 1994; Van Jaarsveld et al., 1999). The minerals almandine, grossular, kyanite, pumpellyite, spodumene, augite, illite, and anorthite contain MgO, with kyanite, augite and anorthite having a relatively high content. It is undesirable for cement to contain more than 5 wt.% MgO, but it is still unknown what effect MgO has on geopolymerisation.

Six minerals — andalusite, kyanite, spodumene, lepidolite, illite and hydroxyapophyllite — show a substantial content of K_2O. The minerals sodalite and heulandite contain a significant amount of Na_2O. In concrete, it is undesirable to have a substantial content of alkali metals owing to alkali activation which causes subsequent stresses. In geopolymerisation, the dissolution reaction and polycondensation steps involve alkali metals, which implies that the alkali metal content of reacting minerals could have a significant effect on strength development.

The XRD patterns of the minerals in Table 1 show varying degrees of crystallisation and contamination. The minerals sillimanite, lepidolite, hydroxyapophyllite, and stilbite have clear patterns with well matched peak positions and peak intensities which means they are pure and highly crystallised minerals. The XRD patterns of the almandine, spodumene, and sodalite samples showed well crystallised minerals, but there were also some weak unknown peaks caused by impurities. The crystallised grossular and kyanite samples showed paragenesis of quartz and zinnwaldite, respectively. The XRD patterns of andalusite, illite, and heulandite had some noise, which indicates that these samples were partly polycrystallised and, moreover, andalusite is the paragenesis of margarite and muscovite, while illite and heulandite are the paragenesis of quartz. A high degree of noise was present in the XRD patterns of celsian, pumpellyite, augite and anorthite, which were partly amorphous and impure. Pumpellyite is the paragenesis of sepiolite, with augite containing ephesite and anorthite containing augite.

4. Extent of dissolution of minerals in alkaline medium

The behaviour of alumino-silicate materials in alkaline solution has been researched extensively (Dent Glasser, 1982; Dent Glasser and Harvey, 1984a,b; McCormick et al., 1989a,b,c; Hendricks et al., 1991; Gasteiger et al., 1992; Antonić et al., 1993, 1994; Devidal et al., 1994; Swaddle et al., 1994). However, all these studies dealt with either pure aluminates, silicates or alumino-silicates and were mostly related to the synthesis of zeolite. There have been some studies on the dissolution and gelatinisation of natural Al–Si minerals in acid medium (Deer et al., 1992). In contrast, little has been done on the reactivity of natural minerals in alkaline medium, mainly as a result of their comparatively lower solubility in alkaline medium than in acid medium.

As stated before, the process of geopolymerisation starts with the dissolution of Al and Si from Al–Si materials in alkaline solution as hydrated reaction products with NaOH or KOH, hence forming the $[M_x(AlO_2)_y(SiO_2)_z \cdot n\mathrm{MOH} \cdot m\mathrm{H_2O}]$ gel. Subsequently, after a short time setting proceeds, with the gel hardening into geopolymers. Consequently, an understanding of the extent of dissolution of natural Al–Si minerals is imperative for an understanding of geopolymerisation reactions.

Table 2 gives the extent of dissolution data of all 16 minerals in terms of the concentration of Al or Si in 20 ml of solution after 5 h of contact with 0.50 g of mineral. The alkaline solutions contained NaOH or KOH at concentrations of 2, 5, and 10 N. The following general trends can be observed from Table 2:

1. Minerals have a higher extent of dissolution with increasing concentrations of alkaline solution.
2. Minerals show a higher extent of dissolution in the NaOH than in the KOH solution, except for sodalite.
3. The concentrations of Si are higher than the corresponding Al, which could be caused partly by the higher content of Si than Al in the minerals, but also by the higher intrinsic extent of dissolution of Si than Al.

Table 2
The extent of dissolution of Si and Al from minerals in NaOH and KOH solutions

Mineral	2 N NaOH		2 N KOH		5 N NaOH		5 N KOH		10 N NaOH		10 N KOH	
	Si (ppm)	Al (ppm)	Si (ppm)	Al (ppm)	Si (ppm)	Al (ppm)	Si (ppm)	Al (ppm)	Si (ppm)	Al (ppm)	Si (ppm)	Al (ppm)
Almandine	59.2	39.6	62.3	39.8	51	34.2	59	36	69.5	44.75	65	41.75
Grossular	60.6	1.5	50.1	1.82	66	2.02	29	1.4	231	3.05	189.5	3.1
Sillimanite	21.1	27.4	17	23.4	23.4	28.4	23.4	26.4	33.75	33.8	39.85	34.65
Andalusite	31.5	33.3	30.2	32.6	31.2	33.2	34	33.6	42.5	43.75	37.05	39.25
Kyanite	22.6	20.9	21.1	20.3	26.4	24.4	24.8	21.6	32.5	30.2	29.85	28.15
Pumpellyite	30.6	14.9	31.1	14.5	19.8	11	29.4	13.68	41.3	20.85	38	18.75
Spodumene	34.2	20.2	29.6	17.5	39.4	23.2	36.4	19.8	54	31.95	45.45	25.75
Augite	59.3	19.8	53.1	20.9	164.8	74.4	83.4	38	215.5	133	236.5	135.5
Lepidolite	36.8	25.1	32.5	22.5	34.4	24.4	37	24.2	42.2	29.35	37.25	27
Illite	42.2	19.8	42	15.8	52	23.4	47	16.56	76	30.6	72.5	29
Celsian	78	62.7	65.8	56.6	78.8	68.2	81.4	63.8	157.5	121	119	97
Sodalite	68.5	13.6	82.1	38	101	37.2	141.2	41.2	78	88.5	301	246
Hydroxyapo-phyllite	58.4	1.28	49.7	1.42	135	2.3	40.8	1.02	140	1.5	107.5	3
Stilbite	116	45.9	98.7	32.9	122.8	44.4	124	44	615	201.5	491	165
Heulandite	127	45.8	94.8	35	141.4	51.6	75	28.4	293	105	216	82.5
Anorthite	86.2	36.2	69.5	29	79.6	36.6	71.2	30	156	73	131	61.5

4. The correlation coefficient between the extents of dissolution of Al and Si is 0.93. Therefore, Si and Al appear to have synchro-dissolution behaviour in alkaline solution, which means that Si and Al could dissolve from the mineral surface in some linked form.
5. Minerals with framework structure possess a higher extent of dissolution than di-, ortho-, ring, chain, and sheet structures in both NaOH and KOH solutions.

Normally, the possible chemical process for the dissolution of Al–Si minerals and silicates under strongly alkaline conditions can be expressed as the following reaction schemes (Babushkin et al., 1985; McCormick et al., 1989b) (M represents the Na or K)

$$\text{Al–Si solid particle} + OH^-(aq) \Leftrightarrow \underset{\text{monomer}}{Al(OH)_4^-} + \underset{\text{monomer}}{{}^-OSi(OH)_3} \tag{3}$$

$$^-OSi(OH)_3 + OH^- \Leftrightarrow {}^-OSi(OH)_2O^- + H_2O \tag{4}$$

$$^-OSi(OH)_2O^- + OH^- \Leftrightarrow {}^-O\overset{O^-}{\overset{|}{Si}}(OH)O^- + H_2O \tag{5}$$

$$M^+ + \underset{\text{monomer}}{{}^-OSi(OH)_3} \Leftrightarrow \underset{\text{monomer}}{M^{+-}OSi(OH)_3} \tag{6}$$

$$2M^+ + \underset{\text{monomer}}{{}^-OSi(OH)_2O^-} \Leftrightarrow \underset{\text{monomer}}{M^{+-}OSi(OH)_2O^{-+}M} \tag{7}$$

$$3M^+ + \underset{\text{monomer}}{{}^-O\overset{O^-}{\overset{|}{Si}}(OH)O^-} \Leftrightarrow \underset{\text{monomer}}{M^{+-}O\overset{O^{-+}M}{\overset{|}{Si}}(OH)O^{-+}M} \tag{8}$$

$$M^+ + \underset{\text{monomer}}{Al(OH)_4^-} + OH^- \Leftrightarrow \underset{\text{monomer}}{M^{+-}OAl(OH)_3^-} + H_2O \tag{9}$$

$$\underset{\text{monomer}}{{}^-OSi(OH)_3} + \underset{\text{monomer}}{M^{+-}OSi(OH)_3} + M^+ \Leftrightarrow \underset{\text{dimer}}{M^{+-}OSi(OH)_2\text{–O–}Si(OH)_3} + MOH \tag{10}$$

$$\underset{\text{monomer}}{{}^-OSi(OH)_2O^-} + \underset{\text{monomer}}{M^{+-}OSi(OH)_3} + M^+ \Leftrightarrow \overset{\text{dimer}}{M^{+-}OSi(OH)_2\text{–O–}Si(OH)_2O^-} + MOH \tag{11}$$

$$\underset{\text{monomer}}{{}^-O\overset{O^-}{\overset{|}{Si}}(OH)O^-} + \underset{\text{monomer}}{M^{+-}OSi(OH)_3} + M^+ \Leftrightarrow \underset{\text{dimer}}{M^{+-}O\overset{O^-}{\overset{|}{Si}}(OH)\text{-O-}Si(OH)_2O^-} + MOH \tag{12}$$

$$2\ \text{Silicate monomer}^- + 2\ \text{Silicate dimer}^- + 2M^+ \Leftrightarrow M^{+-}\text{cyclic trimer} + M^{+-}\text{linear trimer} + 2OH^- \tag{13}$$

As the concentrations of Al and Si in the present study are quite low, mainly these 11 reactions occurred (Babushkin et al., 1985; McCormick et al., 1989b). With concentrated silicate anion addition, the tetramer, pentamer, hexamer, octamer, nonamer, and their compounds will appear (Hendricks et al., 1991). The dissolution reaction (3), for a fixed particle size, is a function of MOH concentration, the structure and the surface properties of the minerals. As the minerals chosen in this paper covered a wide range of structures, compositions, and paragenesis, the factors, which are expected to affect reaction (3), are very complex. In a simplified conceptualisation only the effect of MOH concentration will be discussed below.

From the 11 reactions given above, it can be seen that increasing the concentration of alkaline solution favours all reactions (3) to (9) shifting to the right hand side. Eqs. (3)–(5) are chemical hydration reactions, where the OH^- anions react with the Al–Si solid surface to form $Al(OH)_4^-$, $^-OSi(OH)_3$, divalent orthosilicic acid and trivalent orthosilicic acid ions. Reactions (6) to (9) are physical electrostatic reactions, where the alkali metal cation M^+ reacts with $Al(OH)_4^-$, $^-OSi(OH)_3$, divalent orthosilicic acid and trivalent orthosilicic acid ions to balance Coulombic electrostatic repellence. Reactions (10) to (13) are cation–anion pair condensation interactions based on Coulombic electrostatic attraction. In reactions (9) to (13), the M^+ cation reacts with $Al(OH)_4^-$ and species of orthosilicic acid ions to form ion pairs of $M^{+-}Al(OH)_4$ monomer and silicate monomer, dimer and trimer ions, which reduce the amount of free $Al(OH)_4^-$ and the species of orthosilicic acid ions, therefore shifting reactions (3) to (5) to the right hand side. According to Dent Glasser and Harvey (1984a) there is no cation–anion pair reaction directly on $Al(OH)_4^-$ tetrahedra, which limits the dissolution of Al, so that the concentration of Al is always lower than the corresponding Si concentration of Si.

Reactions (6) to (13) suggest that the alkali-metal cation affects the extent of dissolution of an alumino-silicate. As Na^+ and K^+ have the same electric charge, their different effects are a result of their different ionic sizes. It has been shown that

Table 3
ANOVA for Si concentration in solution vs. $\%SiO_2$ in minerals
Pooled standard deviation (SD) = 2.304.
Individual 95% confidence intervals for mean based on pooled SD.
SS = Sum of squares; MS = mean squares; *df* = degrees of freedom; *P* value = error probability; *F* = *F*-statistic.

Source	*df*	SS	MS	*F*	*P*
Structure	3	90.28	30.09	5.67	0.004
Error	26	138	5.31		
Total	29	228.29			

Level	*N*	Mean	SD
1 (ortho-, di-, ring)	12	1.604	1.331
2 (chain)	4	2.937	2.385
3 (sheet)	4	1.041	0.28
4 (framework)	10	5.273	3.329

Table 4
ANOVA for Al concentration in solution vs. $\%Al_2O_3$ in minerals
Individual 95% confidence intervals for mean based on pooled SD.
Pooled SD = 2.646.
SS = Sum of squares; MS = mean of squares; *df* = degrees of freedom; *P* value = error probability; *F* = *F*-statistic.

Source	*df*	SS	MS	*F*	*P*
Structure	3	226.98	75.66	10.8	0.0001
Error	26	182.1	7		
Total	29	409.08			

Level	*N*	Mean	SD
1 (ortho-, di-, ring)	12	1.263	0.629
2 (chain)	4	5.042	4.57
3 (sheet)	4	1.233	0.289
4 (framework)	10	7.225	3.572

cation–anion pair interaction becomes less significant as the cation size increases. The cation with the smaller size favours the ion-pair reaction with the smaller silicate oligomers, such as silicate monomers, dimers and trimers (McCormick et al., 1989a; Hendricks et al., 1991; Swaddle et al., 1994). Thus, we can expect that Na^+ with the smaller size will be more active in reactions (6) to (13) than K^+, which should result in a higher extent of dissolution of minerals in the NaOH solution (as shown by Table 2). The fact that the sodalite structure is stabilised by sodium but not by potassium may be the reason why sodalite, in contrast with the other minerals, shows a higher extent of dissolution in KOH than in NaOH solution.

A one-way analysis of variance (ANOVA) was conducted on the extent of dissolution data in Table 2 for the different mineral structures in order to determine whether symmetry and structure have a statistically significant effect. Tables 3 and 4 show that the framework structure has a higher extent of dissolution than other structures for both Si and Al, with chain structures being the next highest. The order of the extent of dissolution of the other structures is less clear. The calculated correlation coefficient between the extent of dissolution of Si and Al is 0.93, which suggests that Si and Al are synchro-dissolving from the solid surface.

5. Compressive strength of geopolymers

Table 5 gives the compressive strength of the geopolymers formed from natural Al–Si minerals in NaOH and KOH conditions. A comparison of Tables 2 and 5 shows that some minerals with a higher extent of dissolution such as sodalite and stilbite developed higher than average compressive strength after geopolymerisation. Minerals with a low extent of dissolution, such as grossular and sillimanite do not reveal this relationship, which is indicative of the complexity of these reactions. It is significant that all 15 minerals demonstrate higher compressive strengths after geopolymerisation in

Table 5
The compressive strength of geopolymers formed from Al–Si minerals

Minerals	Compressive strength (MPa)	Compressive strength (MPa)
Alkali	KOH	NaOH
Almandine	10.3	8.5
Grossular	16.7	14.5
Sillimanite	12.7	6.5
Andalusite	11.1	8.8
Kyanite	6.8	6.3
Pumpellyite	10.8	8.8
Spodumene	13.1	5
Augite	6.7	5
Lepidolite	4.3	2.5
Illite	7.1	5.8
Celsian	9.7	8.7
Sodalite	15	10.3
Stilbite	18.9	14.2
Heulandite	7.4	5.6
Anorthite	14.4	6

KOH than in NaOH, despite the higher extent of dissolution in NaOH than in KOH. When KOH was used, the mean compressive strength of all minerals was 11 MPa, which was 42% higher than for NaOH.

It can be expected that the compressive strength developed after geopolymerisation is a highly non-linear function of numerous variables. In order to identify such variables and to quantify the relative importance of these variables, a linear multi-variable regression analysis was performed using the following variables: (a) The %SiO_2, %Al_2O_3, %CaO, %K_2O, %MgO, %Na_2O, and molar Si–Al in the original mineral, the Mohs hardness, the density (g/cm^3) according to Table 1; (b) crystallographic symmetry, where Cubic = 3, Monoclinic = 2, Orthorhombic = 1, Triclinic = 0; (c) type of alkali, where NaOH = 1 and KaOH = 2; (d) extent of dissolution of Si and Al in 10 N alkaline solution (ppm); (e) the molar Si/Al ratio in a 10 N alkaline solution during

Table 6
Correlation coefficients for a linear multi-variable regression analysis between various factors and compressive strength

Factors	SiO_2(s)	Al_2O_3 (s)	CaO (s)	K_2O (s)	MgO (s)	Na_2O (s)	Hardness	Density
Compressive strength	−0.084	−0.256	0.482	−0.530	−0.176	0.146	0.266	−0.101

Factors	Symmetry	Molar Si/Al (s)	NaOH	KOH	Si (ppm)	Al (ppm)	Molar Si/Al (l) ratio
Compressive strength	−0.193	0.420	−0.406	0.406	0.475	0.270	0.413

Table 7
Statistical analysis on coefficients in regression Eq. (14)
$S = 2.909$, $R(\mathrm{Sq}) = 53.7\%$, $R(\mathrm{Sq})_{adj} = 48.3\%$.
P = Error probability, T = T-statistic.

Predictor	Coefficient	SD	T	P
Constant	14.131	1.858	7.61	0
Alkaline used	−3.289	1.063	−3.1	0.005
K_2O	−0.5734	0.2064	−2.78	0.01
Si	0.009366	0.004167	2.25	0.033

dissolution tests. The correlation coefficients between these factors and the compressive strength are shown in Table 6. Despite the fact that a linear correlation is inappropriate as an accurate predictor, it at least provides some guidance to the interpretation of this complex system.

Evidently, factors such as the %CaO, %K_2O and the molar Si–Al in the original mineral, the type of alkali, the extent of dissolution of Si and the molar Si/Al ratio in solution during dissolution tests have a significant correlation with compressive strength. Of these factors, the %CaO, the molar Si–Al in the original mineral, the use of KOH, the extent of dissolution of Si and the molar Si/Al ratio in solution show a positive correlation, while the %K_2O and the use of NaOH correlate negatively with strength. It is worth noting that the hardness of the original minerals, which gives an indication of the original strength, has a positive correlation with the ultimate strength, but it is not as significant as the other variables mentioned above. This suggests that the geopolymeric matrices were are not merely the products of different mineral particles acting as fillers or aggregate in a stabilised gel formed from the dissolution of kaolinite in the presence of sodium silicate. Instead, the significance of the molar Si/Al ratio during the alkaline dissolution of the individual minerals indicates that compressive strength is the result of complex reactions between the mineral surface, kaolinite and the concentrated alkaline sodium silicate solution. After geopolymerisation, the undissolved particles remain bonded in the matrix, so that the hardness of the minerals correlates positively with final compressive strength, as expected. By ‘‘forward selection’’, the following three factors were identified as having a significant effect on strength: (i) the type of alkali, (ii) %K_2O in the mineral, and (iii) ppm Si in solution. All three of these predictors are at the 95% level of significance. Since it has already been demonstrated above that the mineral

Table 8
ANOVA on coefficients in regression Eq. (14)
df = Degrees of freedom; SS = sum of squares; MS = mean of squares; P value = error probability; F = F-statistic.

Source	df	SS	MS	F	P
Regression	3	254.742	84.914	10.03	0
Residual error	26	220.019	8.462		
Total	29	474.762			

structure affects the extent of dissolution of Si, it can be argued that structure affects strength indirectly as well. Eq. (14) gives the regression expression.

$$\text{Compressive Strength (MPa)} = 14.1 - 3.29 \text{ alkaline used} - 0.573 \text{ K}_2\text{O} + 0.00937 \text{ Si ppm} \quad (14)$$

Tables 7 and 8 give a statistical analysis and ANOVA on the regression equation. It becomes apparent that compressive strength cannot be expressed as a simplified function of these variables.

6. Mechanistic considerations

In geopolymerisation, the weight ratio of alumino-silicate powder to alkaline solution is very high, usually between 3.0 and 5.5 (Palomo et al., 1992; Van Jaarsveld et al., 1998). Once the alumino-silicate powder is mixed with alkaline solution, a paste forms which quickly transforms into hard geopolymers. In such a situation, there is not sufficient time and space for the gel or paste to grow into a well crystallised structure such as in the case of zeolite formation.

Figs. 1 and 2 show the XRD patterns of unreacted stilbite and celsian, respectively. Figs. 3 and 4 show the XRD patterns of geopolymeric matrices formed by stilbite/kaolinite and celsian/kaolinite, respectively. A comparison of Figs. 1 and 3, as well as Figs. 2 and 4, shows that after geopolymerisation all main characteristic peaks of both Al–Si minerals and kaolinite still remained, but decreased slightly in intensities.

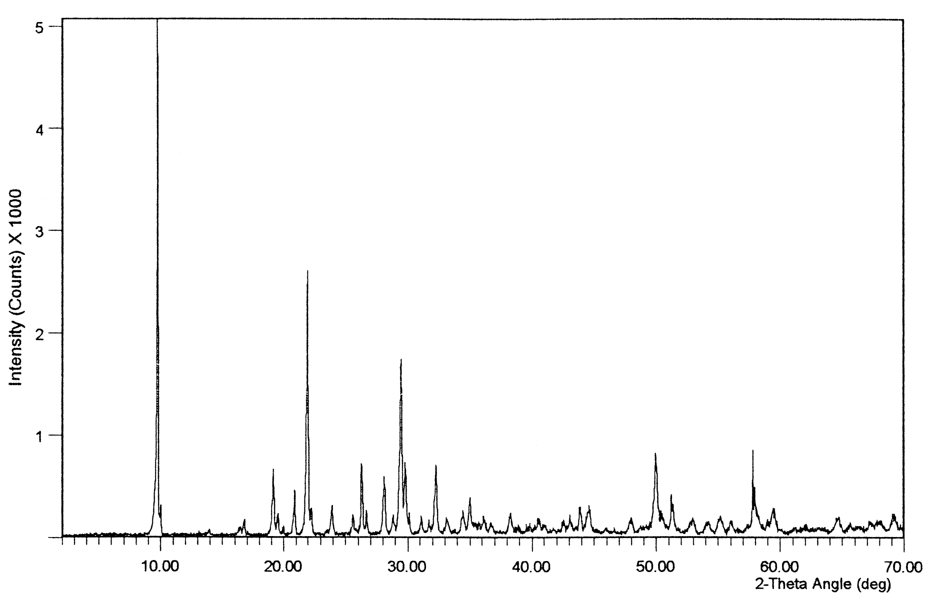

Fig. 1. The XRD pattern of stilbite.

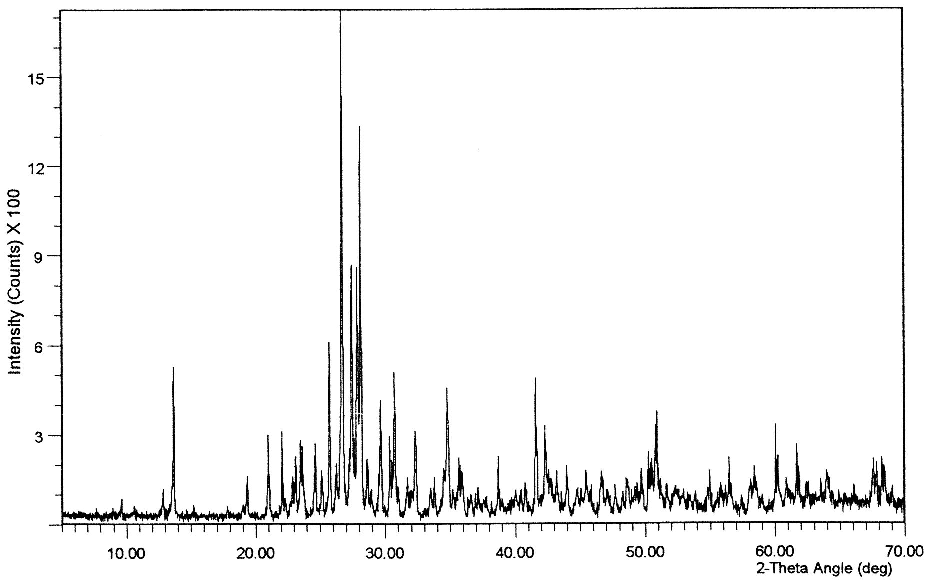

Fig. 2. The XRD pattern of celsian.

This suggests that the stilbite, celsian and kaolinite did not dissolve totally into the gel phase. However, there were no new peaks, which means that no new major crystalline

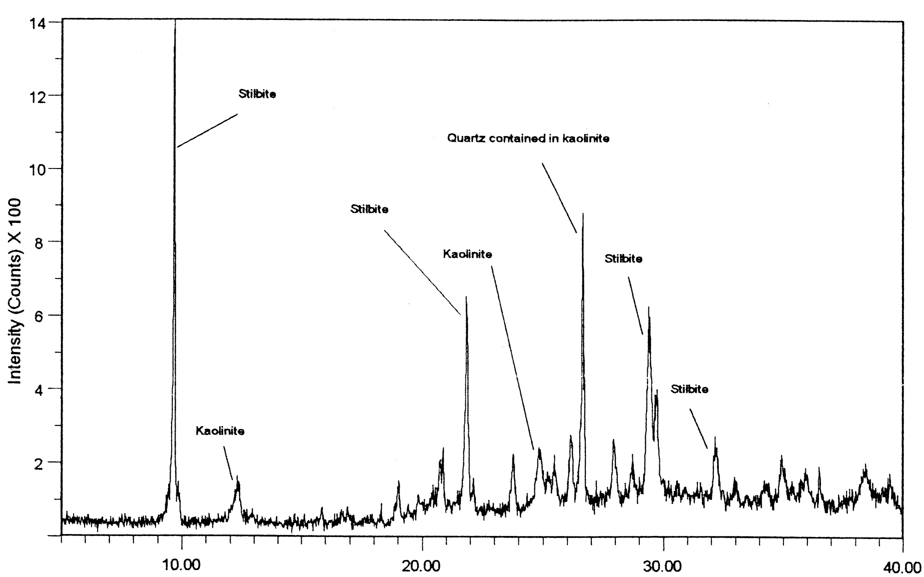

Fig. 3. The XRD pattern of geopolymer formed by stilbite and kaolinite.

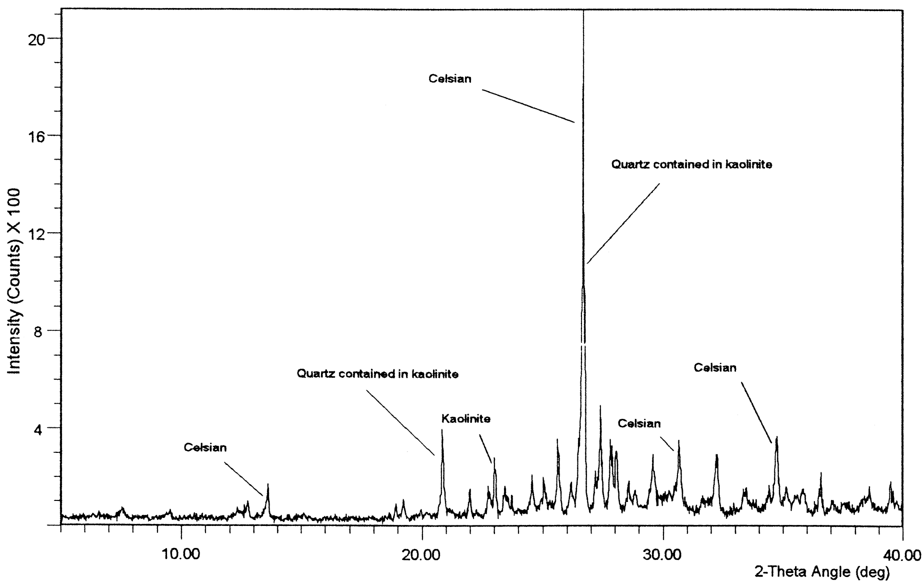

Fig. 4. The XRD pattern of geopolymer formed by celsian and kaolinite.

phases formed. Similar to the observation by Van Jaarsveld et al. (1999), the baseline broadened between 20 and 40° 2θ, which is indicative of an increased amorphicity. Although not shown here, electron diffraction analysis has shown that the formed geopolymer indeed consists of a number of amorphous and poly-crystalline phases.

With the shorter setting and hardening time, geopolymers are formed with tightly packed polycrystalline structure so as to give better mechanical properties than zeolite which have lower density and cage-like crystalline structure. By taking these differences between zeolites and geopolymers into account the following reaction scheme is proposed for the polycondensation process of geopolymerisation from minerals:

$$\text{Al-Si materials (s)} + \text{MOH (aq)} + \text{Na}_2\text{SiO}_3 \text{ (s or aq)} \quad \downarrow \tag{15}$$

$$\text{Al-Si material (s)} + [\text{M}_z(\text{AlO}_2)_x(\text{SiO}_2)_y \cdot n\text{MOH} \cdot m\text{H}_2\text{O}] \text{ gel} \quad \downarrow \tag{16}$$

$$\downarrow$$

$$\text{Al-Si material (s)} \ [\text{M}_a((\text{AlO}_2)_a(\text{SiO}_2)_b) \cdot n\text{MOH} \cdot m\ \text{H}_2\text{O}] \tag{17}$$

Geopolymers with amorphous structure

In reactions (15) and (16), the amount of Al–Si material(s) used depends on the particle size, the extent of dissolution of Al–Si materials and the concentration of the alkaline solution. With finer particle sizes (< 0.5 μm) and hence higher extent of dissolution, comparatively lower ratios of alumino-silicate powder/alkaline solution could be used, as most alumino-silicate particles could then be dissolved as a gel. In most cases, however, alumino-silicate particles cannot be converted totally from the solid phase to the gel phase. Undissolved alumino-silicate solids contained in a geopolymer can behave as reinforcement of the matrix (Palomo et al., 1992). In the present research neither of the 15 minerals dissolved extensively, because their characteristic crystalline peaks could still be detected by XRD after geopolymerisation.

The formation of $[M_z(AlO_2)_x(SiO_2)_y \cdot MOH \cdot H_2O]$ gel, which essentially relies on the extent of dissolution of alumino-silicate materials, is a dominant step in geopolymerisation. Alumino-silicate solids react with MOH solution and form a gel layer on their surfaces. It is proposed that the gel then diffuses outward from the particle surface into larger interstitial spaces between the particles with precipitation of gel and concurrent dissolution of new solid. When the gel phase hardens, the separate alumino-silicate particles are therefore bound together by the gel which acts as a binder.

The gel phase is formed by dissolution from the surfaces of the alumino-silicate listed in Table 1 as well as the added kaolinite. For the purpose of this discussion, the gel is classified in terms of its origin as gel(kao) and gel(Al–Si). It is proposed that the ratio of gel(kao)/gel(Al–Si) depends on the relative extent of dissolution of kaolinite and the Al–Si minerals. Although kaolinite has a much finer particle size (70% < 2.0 μm) than the other Al–Si minerals, the contribution of the Al–Si minerals to the gel phase is still important. A separate experiment was conducted on the composition of the gel phase for the stilbite–kaolinite system with the result showing gel(kao)/gel(stilbite) = 1:1.33. This significant contribution by stilbite to the gel phase could be due to recondensation of the gel, which stimulates further dissolution of the Al–Si minerals. When the gel(kao)/gel(Al–Si) ratio becomes very low, it has been observed that the resulting geopolymers appear cracked, which indicates that the gel formed mainly from dissolution of the Al–Si mineral, but that this gel is not a sufficiently strong binder. On the other hand, if the gel(kao)/gel(Al–Si) ratio is very high, such as for the mineral lepidolite, which has a low extent of dissolution and hence low gel formation, the resulting geopolymer demonstrates low compressive strength. A possible reason for this is the poor wettability between the gel(kao) and the lepidolite solid surface.

In geopolymerisation, high concentrations of silicate are used, especially when sodium silicate is added. Hence, stronger ion-pair formation is expected, which will result in more long chain silicate oligomers as well as Al–O–Si complexes, i.e., geopolymer precursors (McCormick et al., 1989c). In a concentrated alkaline solution of Al and Si, all lengths of silicate could potentially form Al–O–Si complexes. Whereas $Al(OH)_4^-$ does not combine readily with small highly charged silicate oligomers, such as silicate monomers (Dent Glasser and Harvey, 1984a), the more long-chain silicate oligomers exist, the more readily the geopolymer precursors form. This is why the addition of extra Na_2SiO_3 is essential (reaction (15)), as most Al–Si materials cannot supply sufficient Si in alkaline solution to start the geopolymerisation. Na^+, with its smaller size than K^+, displays strong pair formation with the smaller silicate oligomers (such as monomers). Such pairs in turn do not readily pair with another silicate anion (Hendricks et al., 1991), which hinders the further formation of large silicate oligomers. The larger K^+ favours the formation of larger silicate oligomers with which $Al(OH)_4^-$ prefers to bind. Therefore, in KOH solutions more geopolymer precursors exist which result in better setting and stronger compressive strength of the geopolymers than in the case of NaOH.

7. Conclusions

The geopolymerisation behaviour of 16 natural Al–Si minerals was investigated. These minerals were all to some extent soluble in concentrated alkaline solution with a higher extent of dissolution in NaOH than in KOH, except in the case of sodalite. The framework structure showed a higher extent of dissolution than other structures for both Si and Al, with chain structures being the next most soluble. The order of the extent of dissolution of the other structures such as sheet and ring structures was less evident. Silicon and aluminium appeared to be synchro-dissolving from the surface of the minerals, as their extent of dissolution for the different minerals had a high correlation coefficient. Ion pair theory could be used to explain the differences in the extent of dissolution in NaOH and KOH solutions, as well as the increased compressive strength of the geopolymers synthesised in the presence of KOH.

Factors such as the %CaO, %K_2O and the molar Si–Al in the original mineral, the type of alkali, the extent of dissolution of Si and the molar Si/Al ratio in solution had a significant correlation with compressive strength. Stilbite in the presence of KOH showed the highest compressive strength at 18 MPa. Finally, the geopolymerisation results show that natural Al–Si minerals could be a source material for geopolymers. However, it is evident that the reaction mechanisms involved in the dissolution, gel formation, setting and hardening phases are extremely complex and require a great deal of further research. It is still not possible to predict quantitatively whether or not a specific Si–Al mineral will indeed be suitable for geopolymerisation.

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